

Candidate DNN Research Directions

Semester deep neural network project — global workshop venue target

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Candidate DNN Research Directions for a Semester Project (Global Venue Target)

Where this landed

The best fits are multimodal fusion projects where the contribution is architectural or loss-based, running on datasets that are genuinely open and free to grab. Three stand out: EHR-style tabular plus image/signal clinical fusion with a module that stays robust when a modality is missing, multimodal emotion recognition on CMU-MOSEI with a new cross-modal attention or loss design, and self-supervised pretraining for small medical signal datasets like EEG or ECG. All three are doable on Colab or Kaggle's free GPUs and map cleanly onto the four contribution types your course allows.

One thing worth flagging early: skip MIMIC-IV, MIMIC-CXR, and BRSET as your main dataset. They're behind PhysioNet credentialing, which means CITI training plus a signed data use agreement plus an identity review, and that process can take weeks. Go with datasets that need nothing more than a free Kaggle or Hugging Face account instead: PTB-XL for ECG, ISIC or HAM10000 for dermoscopy images with metadata, PAMAP2 for wearable sensor fusion, CMU-MOSEI for multimodal sentiment, MoleculeNet or OGB for graphs, ESC-50 or UrbanSound8K for audio, EuroSAT or BigEarthNet for satellite imagery.

For venues, realistic targets are workshop and short-paper tracks: ICLR, NeurIPS, and ICML all run Tiny Papers or workshop tracks aimed at exactly this kind of project, plus domain-specific ones like ML4RS for remote sensing, the NeurIPS Time Series workshop, and ML4H for medical AI.

Why these directions specifically

Since your course wants the contribution on the deep learning side rather than the data side, the cleanest path for a five-person team without inventing a new architecture from scratch is designing a fusion module, adding a new loss term, and running solid ablations. That's explicitly one of the encouraged contribution types.

Missing-modality robustness and adaptive fusion are a real, current gap, not just something invented to sound novel. A February 2026 benchmark paper on EHR and chest X-ray fusion points out that prior work relied on simple late fusion and never systematically tested what happens when a modality is missing at inference time. A companion preprint makes a similar point about how missing patterns interact with fusion architecture. These are preprints, not peer-reviewed yet, but they reflect a gap that keeps coming up.

Self-supervised pretraining for small labeled datasets, especially in medical imaging, EEG, and ECG, is another active area where the small-data constraint becomes the actual contribution rather than a limitation. A 2023 Nature Scientific Reports paper found masked-autoencoder pretraining held up better than contrastive methods as training set size shrank on small medical CT datasets.

Model compression, distillation, pruning, quantization for edge deployment, is a good fallback because it turns your free-GPU constraint into the point of the paper rather than something to work around, and there are plenty of ready benchmarks to compress against.

Dataset access is the main practical risk. Several well-known medical multimodal datasets are gated behind credentialing, so everything below is chosen to avoid that.

The ten ideas

1. Fusion that survives a missing modality (clinical image or signal plus tabular data)

Multimodal clinical prediction usually beats single-modality models, but that advantage falls apart the moment a modality is missing at test time, which is normal in real clinical settings. That's a genuinely citable open problem right now.

For datasets, use ISIC 2019 (25,331 dermoscopic images across 8 classes, with patient age, sex, and anatomic site as metadata) or HAM10000 (10,015 dermoscopic images across 7 classes, similarly annotated), both downloadable from Kaggle. Pair either with PTB-XL, 21,837 twelve-lead ECG recordings from 18,885 patients, openly licensed under CC BY 4.0 with no credentialing needed, annotated with 71 diagnostic statements grouped into 5 superclasses.

Baseline: separate CNN and MLP branches, combined by simple concatenation. Proposed improvement: a gated cross-attention fusion module trained with modality dropout and a consistency loss, so the fused representation stays useful even when one input is zeroed out at test time. Ablate fusion type, with and without modality dropout, with and without the consistency loss, and performance under each modality dropped individually. This is feasible on a free GPU since the image encoders are small (ResNet-18 or MobileNet scale). Good venues: ML4H, medical AI workshops, or multimodal-learning tracks at ICLR or NeurIPS.

2. Multimodal emotion recognition with a new fusion and loss design (CMU-MOSEI)

CMU-MOSEI has 23,453 annotated utterance segments from 3,228 videos, 1,000 speakers, and 250 topics, over 65 hours of footage, with a standard train/val/test split (16,265 / 1,869 / 4,643). It's heavily benchmarked, so you'll have plenty of published numbers to compare against, and precomputed text, audio, and visual features are available through the CMU MultimodalSDK, which keeps training light. You can extend to MELD or IEMOCAP for cross-dataset generalizability. Text embeddings from an LLM are fine as a preprocessing step here, consistent with your course rules.

Baseline: per-modality encoders concatenated together. Proposed: hierarchical cross-modal attention or an attention-bottleneck fusion, plus an auxiliary modality-alignment or calibrated loss. Ablate attention depth, number of heads, and performance with modalities dropped. Good venues: ACL or EMNLP affective-computing workshops, ICMI, or general multimodal ML workshops.

3. Self-supervised pretraining for small medical signal datasets (EEG or ECG)

This one fits the course's own example almost exactly: plenty of unlabeled medical signal data exists, but labeled data is scarce. Masked-autoencoder and contrastive pretraining are an active 2025 direction here, including masked-reconstruction transformers for seizure prediction and self-supervised frameworks for neurodegenerative disease EEG.

PTB-XL again works well for ECG, since it's open access with no credentialing. For EEG, OpenNeuro or Temple University Hospital corpora are options, though some require registration, so check access before committing to one.

Baseline: a supervised CNN or transformer trained from scratch on a small labeled subset. Proposed: pretrain with masked-signal reconstruction on the unlabeled portion, then fine-tune, with your actual contribution being a domain-specific masking scheme, like masking by lead or frequency band, combined with a small hybrid CNN-transformer encoder. The key experiment is a data-efficiency curve, accuracy versus labeled set size, which directly demonstrates the small-data benefit. Cheap to run since 1D signal models train fast. Good venues: ML4H, ICASSP-adjacent signal workshops, or ICLR/NeurIPS workshop tracks.

4. Wearable sensor fusion for human activity recognition

Human activity recognition from body-worn IMUs is a classic fusion problem, and naive concatenation across sensor locations is still the norm, so placement-aware fusion is genuinely under-explored (recent work like FusionProtoNet targets exactly this).

PAMAP2 from the UCI repository is a solid starting point: 18 activities from 9 subjects wearing three IMUs (wrist, chest, ankle) at 100 Hz plus a heart-rate monitor, 54 attributes per reading, CC BY 4.0 licensed. Add UCI-HAR, WISDM, or OPPORTUNITY for cross-dataset testing.

Baseline: multi-branch CNN with concatenation. Proposed: location-aware attention or graph-based fusion across sensor placements combined with a CNN-transformer temporal encoder. Ablating across sensor subsets also doubles as a missing-sensor robustness test. Trains fast given the small time-series size. Good venues: UbiComp, ISWC, or sensor-fusion workshop tracks.

5. Graph neural networks for molecular property prediction

MoleculeNet and OGB are standard, freely downloadable graph benchmarks, and GNN component design (message-passing variants, fingerprint fusion, global-attention readouts) is still actively researched.

Use MoleculeNet's eight standard splits (BBBP, Tox21, SIDER, ClinTox, ESOL, FreeSolv, Lipophilicity, BACE) or OGB's graph-property datasets.

Baseline: a standard GCN or GIN. Proposed: fuse a GNN branch with a molecular-fingerprint MLP branch through adaptive attention, or add a global-attention readout, plus an auxiliary loss for scaffold generalization. Testing across all eight MoleculeNet splits gives you generalizability experiments for free. Molecular graphs are small, so PyTorch Geometric on a free GPU is plenty. Good venues: AI4Science, AI4Mat, or graph-learning workshop tracks.

6. Self-supervised, multimodal remote sensing

Labeled satellite imagery is limited, so self-supervised learning on abundant unlabeled imagery is a hot area, and fusing multispectral with SAR data adds architectural room. ML4RS has run as a recurring ICLR workshop for several years now, so the venue is well established.

EuroSAT gives you 27,000 labeled 64x64 Sentinel-2 patches across 10 land-use classes, with a benchmark CNN already reaching 98.57% overall accuracy, so you have a strong small baseline to beat. BigEarthNet-MM adds paired SAR and multispectral data for scaling up, and SSL4EO-S12 provides a large unlabeled set for pretraining.

Baseline: supervised CNN on EuroSAT. Proposed: masked-autoencoder pretraining on unlabeled imagery combined with a SAR-optical fusion encoder, with the contribution being the fusion design plus the data-efficiency gain. Stick to EuroSAT or a small BigEarthNet subset to stay within free compute limits, since the full dataset is large. Good venues: ML4RS at ICLR, IGARSS-adjacent workshops, or CV4EO at WACV.

7. Environmental sound classification, either fusion or compression

ESC is a compact, well-benchmarked audio task with two good angles. One is feature fusion: combining log-mel spectrograms, MFCCs, chroma, and spectral contrast into a single CNN-transformer hybrid, which reliably lifts accuracy. The other is going tiny: one recent project compressed an audio transformer down to just over 6,000 parameters, small enough to run on a microcontroller, which is a genuinely publishable direction on its own.

ESC-50 (2,000 clips across 50 classes), UrbanSound8K (8,732 clips), and ESC-10 (400 clips) all come pre-folded for cross-validation.

For the fusion route: baseline is a single-feature CNN, proposed is a stacked-feature CNN-transformer hybrid. For the compression route: distill and quantize/prune an Audio Spectrogram Transformer, reporting accuracy against parameter count and latency, with ablations across compression ratios. Very cheap to run given how small these datasets are. Good venues: DCASE, efficient-AI or on-device ML workshops, ICASSP-adjacent tracks.

8. Model compression for edge deployment

This turns the free-compute constraint into the actual contribution. Recent work notes that systematically improving compression techniques for practical deployment is still under-explored, and task-specific distillation is called out as a gap too.

Any standard benchmark works here since the method is the point: CIFAR-10/100, ESC-50, or PAMAP2 are all fine choices.

Baseline: a large teacher model, ResNet-50 for example. Proposed: a distillation variant (attention-transfer or gradient-guided distillation, say) combined with quantization or pruning down to a MobileNet-scale student, reporting the compression ratio against accuracy and running ablations across ratios and architectures. Very

cheap to run, since small student models train easily on free GPUs, which makes this a safe pick if compute is your binding constraint. Good venues: efficient-AI and edge-AI workshops at ICLR, NeurIPS, or CVPR, and TinyML-adjacent tracks.

9. Network traffic anomaly detection with self-supervised pretraining

The CIC-IDS datasets are large and freely available, and hybrid or ensemble deep models are an active pursuit, with contamination-robust and self-supervised anomaly detection as an open angle.

CIC-IDS2017 has 2,830,743 labeled flow records across 8 CSV files with about 80 CICFlowMeter features, collected over five days in July 2017. CSE-CIC-IDS2018 is bigger still, over 16.2 million flows (about 83% benign, 17% attack), with 75 features. Both are direct downloads from the Canadian Institute for Cybersecurity.

Baseline: an MLP or CNN classifier. Proposed: a self-supervised autoencoder or contrastive pretraining step for anomaly scoring, combined with a hybrid CNN-LSTM temporal branch and an imbalance-aware loss. Ablate across attack types and how much labeled data you use. Subsample for memory, since the full datasets are big, but training is otherwise fast. Good venues: ML-for-security workshops or general ML workshop tracks.

10. Energy load forecasting with a hybrid architecture, outside Sri Lanka

Short-term load forecasting has abundant open data and is a natural way to pivot away from your previous finance-focused project while keeping a forecasting flavor if that's a strength your team wants to reuse. Capturing long-range temporal dependencies alongside nonlinear dynamics is a stated open challenge in recent surveys.

Good datasets: UCI's household electricity data, the ETT family (ETTh1/h2, ETTm1/m2), PJM hourly consumption on Kaggle, ISO-NE data, plus weather covariates from Open-Meteo for a multimodal angle.

Baseline: an LSTM. Proposed: a CNN or TCN combined with a transformer, fused with weather covariates, trained with a probabilistic or quantile loss for uncertainty estimates. Ablate across forecast horizons, covariate sets, and regions for generalizability. Good venues: the NeurIPS Time Series workshop or energy/climate-AI workshops.

What I'd actually pick

If I had to rank them: go with idea 1 or idea 2 first. Both give you a clear, defensible contribution (fusion design plus a loss term), datasets that need no credentialing, plenty of published baselines to compare against, and ablations that fall out naturally from the setup. Between the two, idea 1 (clinical fusion robust to missing modalities) sounds more novel and is backed by a specific, recent open-problem citation, while idea 2 (CMU-MOSEI) is lower risk since the features are precomputed and there's a huge body of prior work to benchmark against.

If compute turns out to be the real bottleneck for your team, fall back to idea 8 (compression) or the tiny-model version of idea 7. Both convert the constraint into the point of the paper and are the safest bets for finishing on time.

For timing: in the first two or three weeks, confirm every dataset you're considering actually downloads and ends before committing to it, reproduce one published baseline number to sanity-check your pipeline, and get the simplest late-fusion baseline running. If your baseline lands within a few points of a published number, keep going; if it doesn't, that's your signal to switch datasets early rather than later. Weeks four through nine are for building the proposed architecture and loss, running ablations, and adding a second dataset for the generalizability experiments your course wants. The last few weeks are for hyperparameter tuning, comparative analysis against baseline and published methods, and writing up for a workshop or short-paper track.

One more thing worth planning for: if your proposed model doesn't beat the baseline after ablations, don't treat that as a dead end. Reframe it as a negative-results or analysis paper instead of abandoning the work. There's an actual venue for this: the ICBINB workshop at ICLR is built specifically around real-world pitfalls and inconclusive results in deep learning.

Things to double check before committing

Dataset access is the biggest risk by far. MIMIC-IV, MIMIC-CXR, and BRSET all require PhysioNet credentialing, CITI training plus a signed data use agreement plus identity review, and while PhysioNet's own FAQ estimates several days to a week, official notices have warned of delays up to 45 days. Don't build your plan around these. The alternatives above (PTB-XL's open tier, ISIC and HAM10000 on Kaggle, PAMAP2 on UCI, EuroSAT) sidestep this entirely.

License-wise, ISIC and HAM10000 are CC BY-NC (non-commercial), which is fine for coursework but worth citing correctly. PTB-XL, EuroSAT, and PAMAP2 are CC BY.

The missing-modality fusion citations above are recent arXiv preprints, not yet peer-reviewed. They reflect a real and widely echoed gap in the field, but cite them as preprints rather than settled findings.

Be a little skeptical if you go the pure tabular-data route: several 2024 and 2025 benchmarks (Grinsztajn et al., McElfresh et al.) find gradient-boosted trees still match or beat deep models on typical tabular data. If you pick a tabular-heavy dataset, frame your contribution around fusion or representation learning rather than "deep learning beats XGBoost," since that claim doesn't hold up well in the literature.

Foundation-model time-series papers often come with strong zero-shot marketing claims. Treat those superlatives cautiously and lean on your own controlled baselines instead.

On compute: the full BigEarthNet and full CIC-IDS datasets may exceed free-tier RAM or GPU limits, so plan to subsample or stream rather than trying to load everything at once.

And on scope: with five people and one semester, resist the urge to add a third modality or a third dataset before your two-modality baseline and first proposed improvement are actually working. The deliverables your course wants, baseline, proposed model, ablations, generalizability, and hyperparameter study, are already a full plate on their own.